MILITARY SPECIFICATION SHEET

INSTALLING AND REMOVAL TOOLS, CONNECTOR ELECTRICAL CONTACT, TYPE III, CLASS 2, COMPOSITION B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the tools described herein shall consist of this document and the latest issue of specification MIL-I-81969.

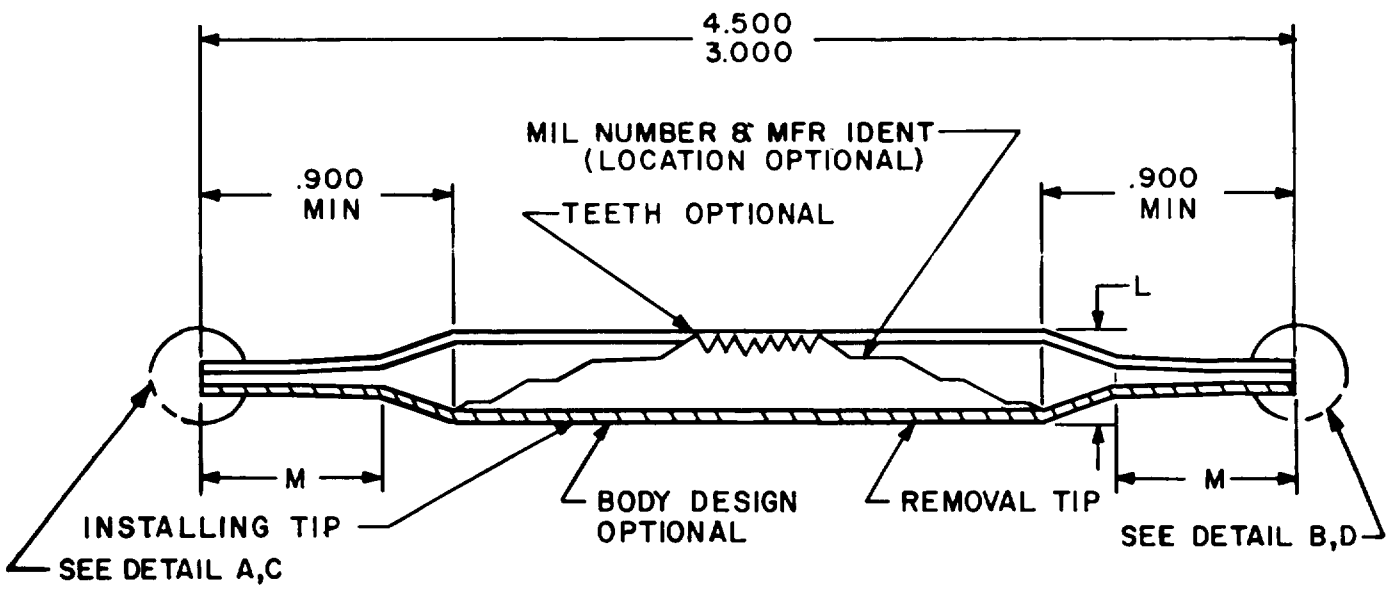


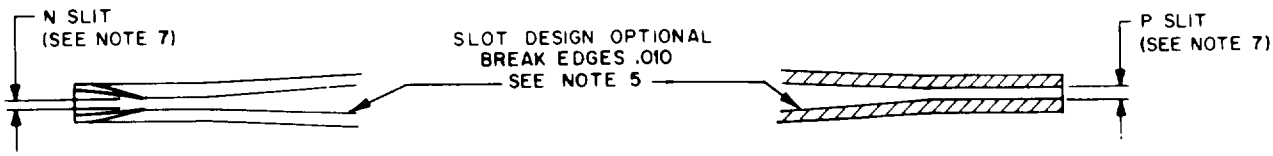
FIGURE 1. Installing and removal tool, -01 and -09.

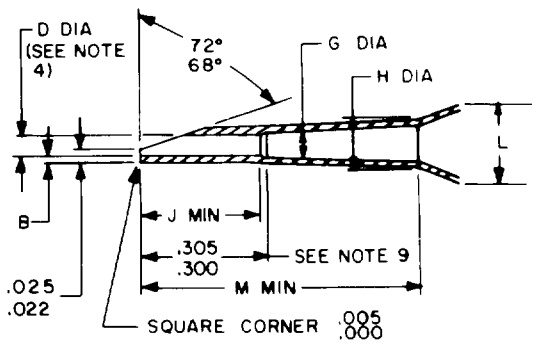
1/ For complete supersession data, see page 8 of this specification.

AMSC N/A FSC 5120

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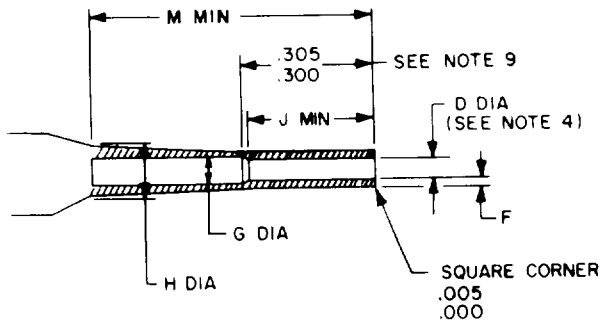
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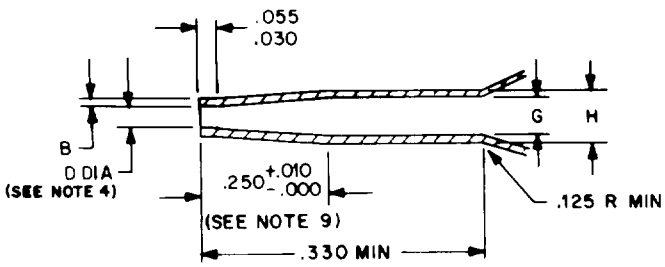
DETAIL A

INSTALLING TIP -01



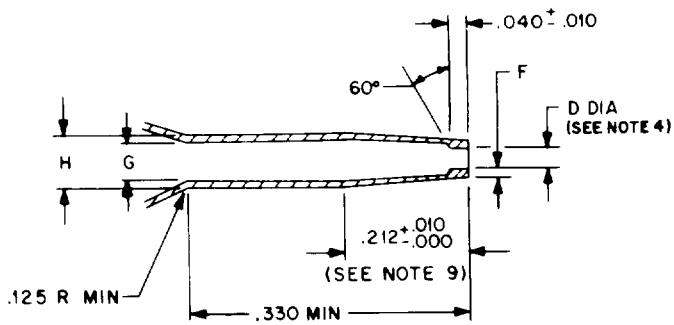
DETAIL B

REMOVAL TIP - 01



DETAIL C

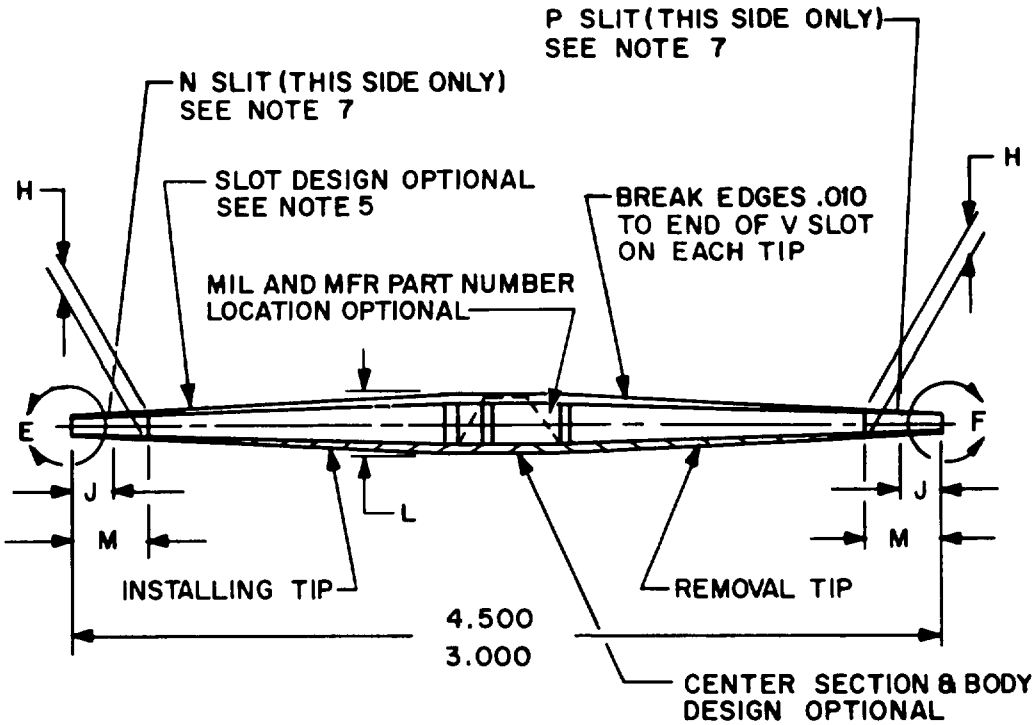
INSTALLING TIP -09

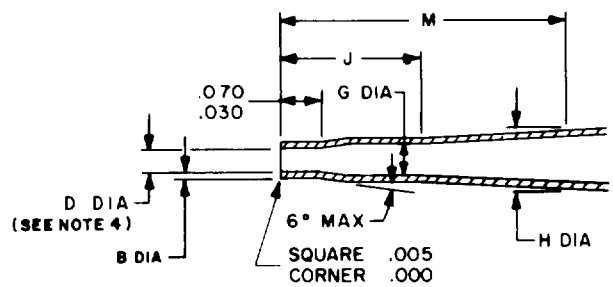


DETAIL D

REMOVAL TIP-09

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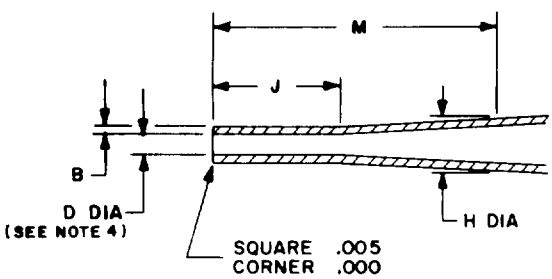




DETAIL E

B DIAMETER MUST BE HELD FOR LENGTH J AS SHOWN

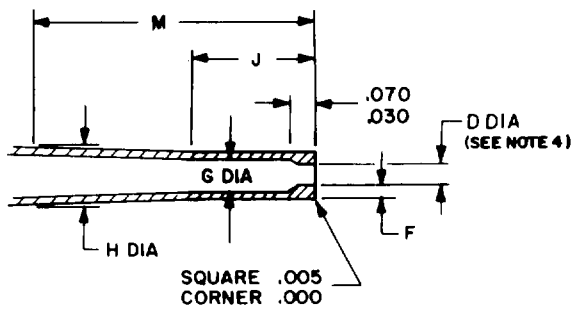
-03, -10 & -11



DETAIL E

OUTER DIAMETER SHALL BE CONSTANT FOR LENGTH J

-04 & -05



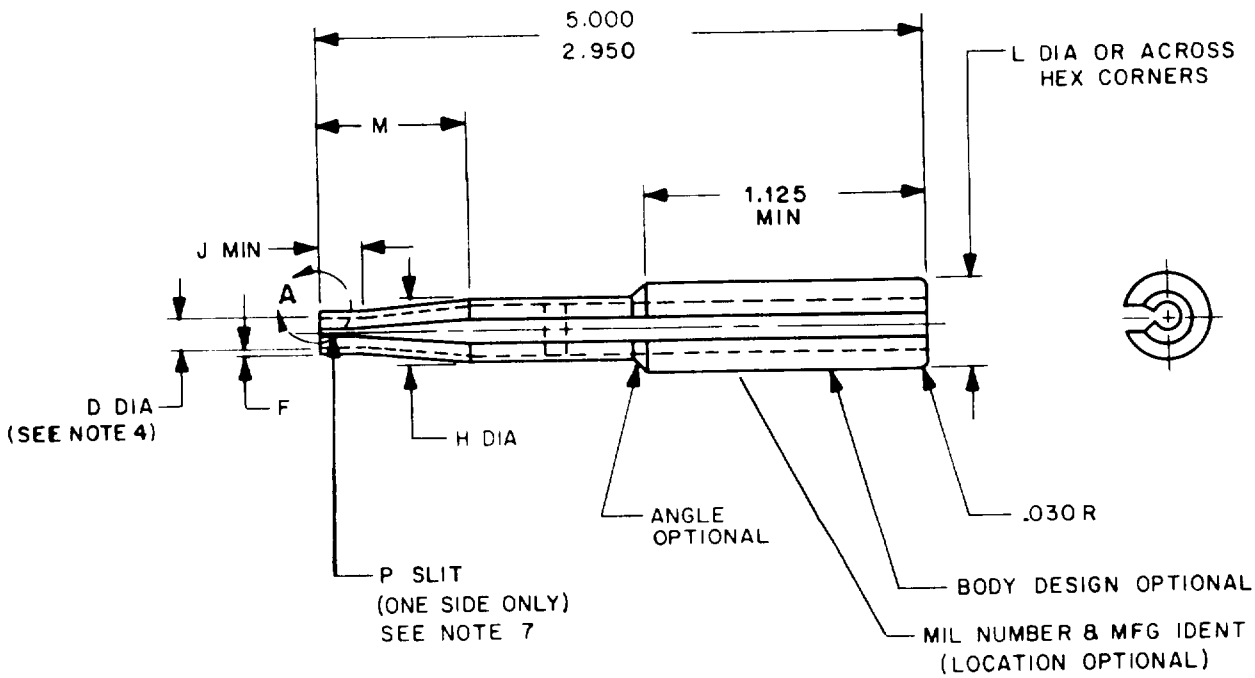
DETAIL F

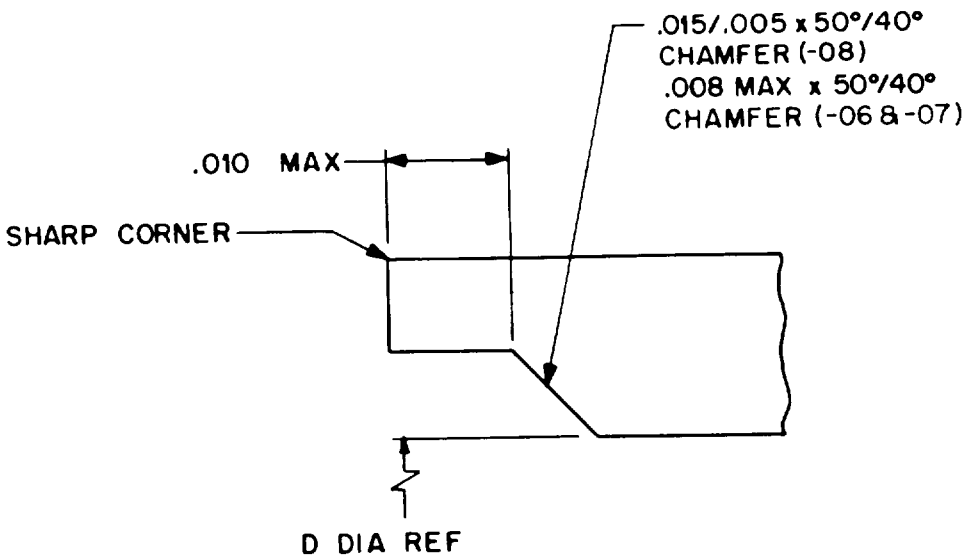
OUTER DIAMETER SHALL BE CONSTANT FOR LENGTH J

-03, -04, -05, -10 & -11

FIGURE 2. Installing and removal tool, -03, -04, -05, -10 and -11.

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DETAIL A

FIGURE 3. Removal tool, -06 thru -08 and -12.

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Inches	mm	Inches	mm	Inches	mm
.005	.13	.071	1.81	.305	7.75
.0070	.180	.072	1.83	.330	8.38
.0075	.190	.076	1.93	.350	8.89
.0080	.200	.078	1.98	.370	9.40
.0085	.216	.080	2.03	.378	9.60
.0090	.23	.083	2.11	.383	9.73
.010	.25	.085	2.16	.390	9.91
.012	.30	.100	2.54	.400	10.16
.0125	.318	.103	2.62	.410	10.41
.013	.33	.104	2.64	.440	11.17
.0135	.343	.105	2.67	.500	12.7
.014	.36	.109	2.77	.537	13.63
.015	.38	.115	2.92	.562	14.27
.0165	.419	.125	3.17	.565	14.35
.0175	.444	.135	3.42	.585	14.86
.018	.46	.150	3.81	.600	15.24
.019	.483	.154	3.91	.640	16.26
.020	.51	.158	4.01	.641	16.27
.022	.559	.160	4.06	.760	19.30
.025	.64	.165	4.19	.765	19.43
.026	.660	.171	4.34	.785	19.94
.027	.686	.175	4.44	.875	22.13
.028	.711	.205	5.21	.900	22.86
.030	.76	.207	5.26	1.000	25.4
.034	.85	.212	5.38	1.125	28.58
.035	.89	.213	5.41	1.150	29.21
.040	1.01	.215	5.47	1.200	30.48
.048	1.22	.231	5.87	1.210	30.73
.050	1.27	.235	6.02	2.000	50.80
.051	1.29	.242	6.15	2.500	63.50
.052	1.32	.250	6.35	2.950	74.93
.054	1.37	.272	6.92	3.000	76.20
.058	1.47	.274	6.96	4.500	114.30
.066	1.67	.278	7.06	5.000	127.00
.069	1.75	.290	7.37		
.070	1.78	.300	7.62		

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TABLE I. Dash numbers and dimensions.

Dash No.	Contact Size	B	D Dia		F	G	H Max 1/	Color Code		J Min.	L Max.	M Min.	N	P
			Installing Tip	Removal Tip				Installing Tip	Removal Tip					
-01	22D	.0085 .0075	.048 .046	.050 .048	.0085 .0075	.058 .054	.077	Green	White	.290	.290	.600	.012 .007	.012 .007
-03	16	.010 .009	.104 .100	.105 .103	.016 .015	.115 .109	.231	Blue	White	.370	.440	1.000	.020 .010	.016 .010
-04	12	.012 .010	.154 .150	.160 .158	.0175 .0165	--	.287	Yellow	White	.370	.440	1.000	.022 .019	.016 .010
-05	10	.012 .010	.207 .205	.215 .213	.018 .017	--	.290	Gray	White	.500	.440	1.200	.026 .022	.026 .022
-06	8	--	--	.278 .272	.020 .017	--	.400	--	Red	.585	.641	1.000	--	.030 .026
-07	4	--	--	.390 .383	.019 .017	--	.537	--	Blue	.585	.875	1.000	--	.034 .030
-08	0	--	--	.565 .562	.028 .027	--	.765	--	Yellow	.785	1.210	1.150	--	.035 .030
-09	22	.008 .007	.052 .051	.049 .046	.0115 .0105	.0605 .0595	.076	Brown	White	--	.350	--	.015 .010	.015 .010
-10	20	.009 .008	.070 .066	.071 .069	.0135 .0125	.085 .083	.150	Red	Orange	.370	.290	.760	.014 .010	.012 .007
-11	20	.009 .008	.070 .066	.080 .078	.014 .013	.085 .083	.150	Red	White	.370	.290	.760	.014 .010	.012 .007
-12	8	--	--	.278 .274	.022 .021	--	.390	--	Green	.500	.640	1.200	--	.030 .026

1/ H represents the maximum tool O.D. thickness at a minimum distance M.

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NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.000 inch = 25.40 mm.
3. Tips shall be capable of being separated from center section -01 thru -05, -09 thru -11.
4. "D" diameter represents tool tip inner diameter in free state.
5. Slot and body design shall allow easy entry of all applicable wire sizes.
6. Each tool shall be legibly and permanently marked on the handle in accordance with MIL-STD-130.
7. Slit is measured with the tool in a free state.
8. Wires having a finished diameter greater than the crimp barrel O.D. may require the use of a M81969/8 tool for contact removal.
9. This is the point where the "G" dimension begins.
10. Tool O.D. shall increase at a constant slope from the J to the H dimension.

REQUIREMENTS:

Dimensions and configurations: See Figures 1, 2, 3 and Table I.

Material: Delrin or equivalent.

Plating: Not applicable.

Plunger force: Not applicable.

Part Number: M81969/14- (and dash number from Table I).

Application and Supersession Data: These tools shall be used to install and remove contacts from connector as specified in Table II.

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TABLE II.

Size	For Use With MIL-C-55302/68, /69 MIL-C-38999 MIL-C-24308 (Size 22D) MIL-C-83733 (Size 22D) MIL-T-81714 (Size 22D)	For Use With MIL-C-26482 Series II MIL-C-81703 Series III MIL-C-83733 (Except 22D) MIL-C-83723 Series II & III MIL-C-5015 Series 3450 MIL-T-81714 (Except 22D)	Superseded Part Numbers
22D, 22M	M81969/14-01	----	MS27534-22D
22	M81969/14-09 1/	----	----
20	M81969/14-10	M81969/14-11	M81969/14-02 M27509R20 MS27534-20 NAS1664-20 MS3447-20 M83723/31-20
16	M81969/14-03	M81969/14-03	MS3447-16 MS3166-16 MS3167-16 MS27534-16 NAS1664-16 M83723/31-16
12	M81969/14-04	M81969/14-04	MS3447-12 MS3166-12 MS3167-12 MS27534-12 NAS1664-12 M83723/31-12
10	M81969/14-05	----	----
8	M81969/14-12	M81969/14-06	M81969/29-02
4	----	M81969/14-07	M81969/29-03
0	----	M81969/14-08	M81969/29-04

1/ For MIL-C-55302/169, /171 only
The symbol ---- indicates not applicable

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Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - MI, AR, AT, GL
Navy - EC, SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AV, ME
Navy - MC, YD
Air Force - 19

Preparing activity:
Navy - AS
(Project No. 5120-B073)

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